

Generation Interconnection System Impact Study Report

For Queue Project Q22-2012

Sand Springs 138 kV

December 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2012. The Interconnection Customer proposes a 184.6 MW solar photovoltaic facility interconnecting at the Sand Springs 138 kV substation in the Kentucky area. The request was received on February 19, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 510 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$463,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Louisiana 1 / Desoto Pleasant Run 138.0 kV

Contingency: Mittersky / Southaven 138.0 kV

Loading: 102.0% (Rating: 171 MVA)

DFAX: 8.3%

Pre-project loading: 93.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Louisiana 1 / Desoto Pleasant Run 138.0 kV	Reconductor line	\$3,343,000	14% (\$463,000)
	Total allocated to project		\$463,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.